

AIMO Proof Pilot – Problems, Solutions, and Markschemes*

June 2026

Overview

This document accompanies the AIMO Proof Pilot. It contains the six problems used in the competition, each presented with a reference solution and a detailed markscheme. The remainder of this section sets out which problems were chosen, how solutions are scored, the goals of the pilot, and the steps we have taken to finalise the markschemes *before* any evaluation was carried out.

The Problems

We have selected **six problems** from the AIMO3 public¹ problem set. The problems have been adapted slightly for the proof-based format. The six were chosen to span a range of (human) **difficulties** and a range of **topics** (across algebra, combinatorics, geometry, and number theory). Each problem heading lists its six-letter public identifier in brackets (e.g. “DIVALL”), matching the identifiers used elsewhere in the AIMO3 materials, so that outputs and discussion can be cross-referenced unambiguously.

To give a concrete sense of how teams performed on these problems during AIMO3, the table below reports the results from the public leaderboard, where the problems were posed in *answer-only* form. For each problem we show the solve rate across all teams (based on each team’s best submission), alongside the rates among the top-ranked teams (the top 100, 25, 10, and 5 on the public leaderboard).

Problem		AIMO3 solve rate (%)				
#	ID	All teams	Top 100	Top 25	Top 10	Top 5
1	DIVALL	76.3	100	100	100	100
2	ALTORO	81.8	100	100	100	100
3	SINDEX	78.5	100	100	100	100
4	COLOUR	52.1	89	100	100	100
5	POLYGO	50.4	89	96	100	100
6	AMAZIN	1.8	2	0	0	0

It will be interesting to see how teams fare when faced with the task of producing a full proof compared with just having to find the correct answer.

Goals and Structure of the Proof Pilot

The AIMO Proof Pilot is an invite-only competition that explores how well LLM-based systems can produce **mathematical proofs** at a standard closer to the International Mathematical Olympiad (IMO), rather than only producing final numeric answers. Because (natural language) proofs cannot

*To cite these problems, please refer to the citation of the associated Kaggle competition.

¹As with AIMO3, “public” here does not mean the problems are publicly available. It refers to the set used for the public leaderboard; the problems themselves are not released.

be checked automatically (at least not with 100% accuracy), every output requires **human evaluation**, and the purpose of this document is to make that evaluation as transparent, consistent, and reproducible as possible.

The pilot also introduces a deliberately restricted model pool: rather than allowing any open-weight model, it permits only a small set of **fully open-source** models for which intermediate checkpoints, training protocols, and training data are publicly available (such as models from the OLMo family). This makes a meaningful post-mortem possible—when a model performs well on one problem and poorly on another, we have a realistic chance of understanding why.

For each problem we provide a reference solution followed by a markscheme. The markscheme is the authoritative document for grading: the reference solution illustrates one route to a complete answer, but graders score the *model's* argument on its own merits against the criteria, not on its resemblance to ours.

Marking and Alternative Approaches

Olympiad problems typically admit many correct approaches, and we expect model outputs may differ substantially from the reference solutions. The markscheme below is therefore structured around the **key mathematical milestones** of each problem rather than a single fixed write-up. Where a model takes a genuinely different but valid route, graders are instructed to map that route onto the same milestones: an alternative approach that reaches the equivalent decisive step earns the equivalent credit. Where we anticipate common alternative strategies, we note explicitly how they fit into the framework.

Scoring scale (0/1/6/7). We use a simplified four-point scale—0, 1, 6, or 7—rather than the full 0–7 range used at the IMO. This reduces grading to **two binary decisions** and so makes scoring faster, more consistent between graders, and easier to reconcile when graders disagree:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

By collapsing the intermediate marks, the scale avoids the hardest and most contentious judgements of olympiad grading (deciding precisely how many of 7 marks a partial argument is worth), where graders most often disagree.

A stricter bar than the IMO. Two consequences of this structure are worth highlighting. First, our criterion for an **essentially complete** solution is generally *stricter* than the corresponding bar at the IMO: a solution must be very close to a complete solution, and we do not allow for some larger gaps that the IMO may view as sufficiently minor to still receive near-full credit. Second, **partial credit is also generally harsher**: there is no continuum of 2–3 marks, so there is less distinction between levels of significant progress, and often a higher bar to score the 1 mark.

The 6-mark band (minor omissions). Unless a problem's markscheme states otherwise, a solution that is essentially complete but contains **one or more minor errors or omissions** still scores **6**. “Minor” means the gap is small and would be easy to correct without making any structural changes to the solution. We reserve the right, however, to treat an *accumulation* of such gaps as more than minor: where there are many small omissions, their combined effect can amount to a

material hole in the argument, in which case the solution is no longer essentially complete and drops into the partial-progress band (0 or 1). The per-problem markschemes specify, where relevant, which omissions are considered minor and which are major.

Integrity: Markschemes Fixed Before Evaluation

A central commitment of the Proof Pilot is that **the markschemes in this document were drafted in full before any evaluation was conducted**—in particular, before any model outputs were generated in the Kaggle environment. This matters because the credibility of human grading depends on the criteria being fixed in advance, so that scores cannot be adjusted after the fact to favour or disfavour particular solutions.

To make this verifiable rather than merely asserted, we take two steps before the evaluation begins:

1. **Published hash.** We compute a cryptographic hash (SHA-256) of this finalised document and publish it in advance. Any later change to the solutions or markschemes would change the hash, so the published value commits us to exactly this version.
2. **Advance disclosure to Kaggle.** We share the complete document with Kaggle ahead of the evaluation, so that Kaggle can independently confirm that the markschemes existed, and were finalised, before any outputs were produced.

Together these steps allow anyone to verify, after results are released, that the grading criteria were settled before the models were run.

Problems

Problem 1 (DIVALL)

There are $k \geq 3$ different positive integers $a_1, \dots, a_k$ with product P , and sum strictly less than $3 \cdot 10^6$. Let g be the largest positive integer that divides each of $(P/a_i) + 1$ for $1 \leq i \leq k$. Across all possible choices for a_i and $k \geq 3$, what is the maximum possible value of g ?

Problem 2 (ALTORO)

Let ABC be a triangle such that $AB = 26$, $BC = 17$, and $AC = 25$. Let Ω and H be the circumcircle and orthocentre of triangle ABC , respectively. Let the feet of the altitudes from A , B , and C in triangle ABC be D , E , and F , respectively. Let lines EF and BC intersect at X and let AX intersect Ω again at $Q \neq A$. Segment HQ intersects circle DEF at a point P . Suppose the ratio $PQ/PE = a/b$ where a and b are coprime positive integers. What is the value of $a + b$?

Problem 3 (SINDEX)

For a finite, nonempty multiset S of positive integers strictly less than 20, we define its *index* to be the greatest positive integer n such that for every i with $1 \leq i \leq n$ we can partition S into multisets A_i and B_i in such a way that $A_i = i + B_i$. If no such n exists, we define the index to be equal to 0.

Let N denote the maximum index across all choices for S . Across all choices of S with index N , let M denote the smallest possible value for the maximum element in S .

Across all multisets S with index N and maximum element M , what is the smallest possible number of elements of S (counted with multiplicity)?

For two multisets X and Y and a positive integer l , we write $X = l + Y$ if $X = \{l + y : y \in Y\}$ where the right-hand side is interpreted as a multiset.

Problem 4 (COLOUR)

Linda and Sue play a game on a 2025×2025 grid of cells. Initially, every cell is grey. Linda moves first, and the players then alternate turns.

On each of Linda's turns, she chooses one cell and paints it lime, regardless of its current colour. On each of Sue's turns, she chooses an axis-aligned 2×2 block of cells containing at least one grey cell, and paints all four cells scarlet, regardless of their current colours.

After each move, if there are no grey cells left, the game ends immediately. Linda wants to maximise the number of cells whose final colour is lime, and Sue wants to maximise the number of cells whose final colour is scarlet.

- Prove that the game always ends after finitely many moves.
- Suppose both players play optimally. Determine the number of lime cells at the end of the game.

Problem 5 (POLYGO)

A polynomial $P(x)$ is called *good* if it has integer coefficients and:

- It is a monic polynomial of degree 7;
- The absolute values of all of the coefficients of P are odd and strictly less than 2025^2 ;
- The absolute values of all the coefficients of P are roots of P .

How many good polynomials are there?

Problem 6 (AMAZIN)

Let ABC be an acute triangle with $\angle CBA = 75^\circ$, orthocentre H , and circumcircle Ω . Let line AH meet Ω again at $H_A \neq A$. Similarly, let line CH meet Ω again at H_C . Let the circle centred at B with radius BH meet the line BC at X on the opposite side of B from C . Assume that lines XH_A and CH are not parallel, and let them intersect at P . Let lines XH_C and AH intersect at Q .

Let n be an odd positive integer. We call a triangle ABC *n-good* if $\frac{CP}{CH} = \sqrt{n}$ and the length AQ is a positive integer. We say n is *amazing* if there exists an *n-good* triangle where the length H_CQ is also a positive integer.

Let $1 \leq n_1 < n_2 < n_3 < \dots$ be a list of all amazing, odd positive integers.

Let x be the largest real number for which there exists a positive real number y such that for all sufficiently large i , $n_{2i} \geq y \cdot x^i$. For this choice of x , let z be the largest real number such that for all sufficiently large i , $n_{2i} \geq z \cdot x^i$.

It is possible to write $x + 2016z = p + q\sqrt{r}$ where p , q , and r are positive integers with r square-free. What is $p + q + r$?

Problem 1 (DIVALL)

Problem: There are $k \geq 3$ different positive integers $a_1, \dots, a_k$ with product P , and sum strictly less than $3 \cdot 10^6$. Let g be the largest positive integer that divides each of $(P/a_i) + 1$ for $1 \leq i \leq k$. Across all possible choices for a_i and $k \geq 3$, what is the maximum possible value of g ?

Original AIMO3 Formulation:

There are $k \geq 3$ different positive integers $a_1, \dots, a_k$ with product P , and sum strictly less than $3 \cdot 10^6$. Let g be the largest positive integer that divides each of $(P/a_i) + 1$ for $1 \leq i \leq k$. Across all possible choices for a_i and $k \geq 3$, let G be the maximum possible value of g . What is the remainder when G is divided by 10^5 ?

Answer: 98285

Remark: Relative to the AIMO3 version, we removed the modulo 10^5 answer modification as we are not restricted to 5-digit answers in the Proof Pilot.

Solution: We prove that the maximum is $\boxed{998285}$.

Claim 1: g and P are coprime.

Proof. Suppose, for contradiction, that some prime q divides both g and P . Since $P = a_1 a_2 \cdots a_k$, the prime q divides some a_i . Choose an index $j \neq i$, which is possible since $k \geq 3$. Then a_i is one of the factors of P/a_j , so $q \mid P/a_j$. But $q \mid g$, and g divides $(P/a_j) + 1$, so $q \mid (P/a_j) + 1$. Hence q divides both P/a_j and $(P/a_j) + 1$, which is impossible. Thus no prime divides both g and P . $\square$

Claim 2: For $1 \leq i < j \leq k$ we have $g \mid a_j - a_i$.

Proof. Since g divides both $(P/a_i) + 1$ and $(P/a_j) + 1$, it divides their difference. Thus,

$$g \mid ((P/a_i) + 1) - ((P/a_j) + 1) = \frac{P}{a_i a_j} (a_j - a_i).$$

The factor $P/(a_i a_j)$ is a product of some of the a_ℓ , so it divides P . By Claim 1, it is therefore coprime to g . This means that g must divide the factor $(a_j - a_i)$ as claimed. $\square$

It follows from Claim 2 that all the a_i leave the same residue $(\text{mod } g)$, say $0 \leq r < g$. If $g = 1$, then certainly $g < 998285$, so for the upper bound we may assume $g \geq 2$. In this case, Claim 1 tells us $r \neq 0$. WLOG order the numbers so that $a_1 < a_2 < \cdots < a_k$. Since the a_i are distinct positive integers in the same residue class modulo g , we have

$$a_i \geq r + (i - 1)g \quad \text{for } 1 \leq i \leq k.$$

Also, for every i ,

$$0 \equiv (P/a_i) + 1 \equiv r^{k-1} + 1 \pmod{g},$$

so

$$g \mid r^{k-1} + 1. \tag{1}$$

We now split into cases according to k .

Case 1: $k = 3$. From the lower bound on the three a_i ,

$$3 \cdot 10^6 > a_1 + a_2 + a_3 \geq r + (g + r) + (2g + r) = 3(g + r),$$

so

$$g + r < 10^6. \tag{2}$$

By (1), we have $g \mid r^2 + 1$. Write

$$r^2 + 1 = \ell g$$

for some positive integer ℓ .

- If $\ell = 1$, then $g = r^2 + 1$. By (2),

$$r^2 + r + 1 = g + r < 10^6,$$

so $r \leq 999$, and therefore

$$g = r^2 + 1 \leq 999^2 + 1 = 998002 < 998285.$$

- If $\ell = 2$, then

$$g + r = \frac{r^2 + 1}{2} + r = \frac{(r + 1)^2}{2} < 10^6.$$

Hence $r + 1 < 1000\sqrt{2}$, so $r \leq 1413$. Therefore

$$g = \frac{r^2 + 1}{2} \leq \frac{1413^2 + 1}{2} = 998285.$$

This is attainable by setting $r = 1413$, $g = (r^2 + 1)/2$, and

$$a_1 = r, \quad a_2 = g + r, \quad a_3 = 2g + r.$$

which are distinct positive integers. The divisibility conditions then hold since $(P/a_i) + 1 \equiv r^2 + 1 \pmod{g}$ and by construction the sum $3(g + r) < 3 \cdot 10^6$.

- If $\ell = 3$, then no integer r is possible, because squares are congruent to 0 or 1 modulo 3, so $r^2 + 1$ is congruent to 1 or 2 modulo 3 and is never divisible by 3.
- Finally suppose $\ell \geq 4$. It is enough to rule out the possibility that $g > 998285$. If $g > 998285$, then $g \geq 998286$, so (2) gives $r \leq 1713$. But then

$$g = \frac{r^2 + 1}{\ell} \leq \frac{1713^2 + 1}{4} < 998285,$$

which is a contradiction. Thus, in the case $k = 3$, we always have $g \leq 998285$.

Case 2: $k \geq 4$. Using only the four smallest of the a_i , we get

$$3 \cdot 10^6 > a_1 + a_2 + \cdots + a_k \geq a_1 + a_2 + a_3 + a_4 \geq r + (g + r) + (2g + r) + (3g + r).$$

The right-hand side is $6g + 4r$, which is greater than $6g$. Therefore

$$3 \cdot 10^6 > 6g,$$

so $g < 500000 < 998285$. Hence the case $k \geq 4$ cannot improve on the bound found for $k = 3$.

Thus, the maximum possible value for g is $\boxed{998285}$ as claimed.

Markscheme: We follow a two-step process for scoring solutions.

Step 1. Decide whether the solution is essentially complete.

Step 2. If it is not essentially complete, award 1 mark for significant partial progress and 0 marks otherwise. If it is essentially complete, award 7 marks unless there is a minor error or omission, in which case award 6 marks.

Essentially complete solutions (6/7 marks)

A solution is *essentially complete* if, allowing only minor slips of the kind described below, it proves both the upper bound $g \leq 998285$ and a construction attaining $g = 998285$.

An essentially complete solution should receive 6 rather than 7 if it contains a minor error or omission: a gap that is not merely a small arithmetic slip, but which can be repaired without changing the main structure of the solution. Examples include:

(ME1) Proving $\gcd(g, P) = 1$ correctly, and later asserting

$$g \mid \frac{P}{a_i a_j} (a_j - a_i) \implies g \mid a_j - a_i$$

without explicitly saying that $P/(a_i a_j)$ is coprime to g . This is a minor omission only when there is a valid proof of $\gcd(g, P) = 1$ somewhere else in the solution.

(ME2) Omitting the explanation that the common residue r is non-zero, while otherwise using the common-residue argument correctly.

(ME3) Making a fixable arithmetic or off-by-one error in one of the small $k = 3$ subcases, such as mishandling the strict inequality in the $\ell = 2$ case, while the intended bound and final answer are otherwise clear.

(ME4) Giving the correct construction in the $\ell = 2$ case but failing to verify one or more of its required properties, such as the divisibility condition or the sum bound.

The following should *not* generally be penalised:

(NME1) Minor arithmetic slips in the $k \geq 4$ case that still leave a bound comfortably below 998285.

(NME2) Not proving separately that the greatest common divisor of the numbers $(P/a_i) + 1$ in the constructed example has greatest common divisor exactly 998285 (i.e. exclude the case that the gcd could be a larger multiple of this), provided the solution has already proved the global upper bound and has checked that 998285 divides all the relevant numbers in the example.

The following are not essentially complete, even if they contain many correct later calculations:

(NEC1) A solution that needs but does not prove $\gcd(g, P) = 1$.

(NEC2) A solution that assumes the a_i are pairwise coprime, or otherwise adds an assumption not present in the problem.

(NEC3) A solution that considers only $k = 3$ and does not rule out $k \geq 4$.

(NEC4) A solution that gives only the construction, or only an upper bound, but not both.

Partial progress for incomplete solutions (0/1 marks)

Award 1 mark to an incomplete solution if it gives a correct proof of the common-residue result

$$g \mid a_i - a_j \quad \text{for all } i, j,$$

or an equivalent statement forcing all the a_i to lie in a single residue class modulo g . To receive this mark, the solution must include any coprimality argument it requires, such as first proving $\gcd(g, P) = 1$ and then using it to ‘cancel’ $P/(a_i a_j)$ from the divisibility.

Do not award the partial-progress mark for any of the following on its own:

(NPP1) Proving only that $\gcd(g, P) = 1$.

(NPP2) Finding or guessing the example with $g = 998285$ (including verifying the example works).

(NPP3) Any bounds following from the assertion, rather than proof, that the a_i have a common residue modulo g .

(NPP4) Claiming $g \mid a_i - a_j$ from

$$g \mid \frac{P}{a_i a_j} (a_j - a_i)$$

without proving or otherwise justifying the coprimality needed to cancel $P/(a_i a_j)$.

(NPP5) Numerical experimentation, pattern-spotting, checking several examples etc.

Problem 2 (ALTORO)

Problem: Let ABC be a triangle such that $AB = 26$, $BC = 17$, and $AC = 25$. Let Ω and H be the circumcircle and orthocentre of triangle ABC , respectively. Let the feet of the altitudes from A , B , and C in triangle ABC be D , E , and F , respectively. Let lines EF and BC intersect at X and let AX intersect Ω again at $Q \neq A$. Segment HQ intersects circle DEF at a point P . Suppose the ratio $PQ/PE = a/b$ where a and b are coprime positive integers. What is the value of $a + b$?

Original AIMO3 Formulation:

Let ABC be a triangle such that $AB = 26$, $BC = 17$, and $AC = 25$. Let Ω and H be the circumcircle and orthocentre of triangle ABC , respectively. Let the feet of the altitudes from A , B , and C in triangle ABC be D , E , and F , respectively. Let lines EF and BC intersect at X and let AX intersect Ω again at $Q \neq A$. Segment HQ intersects circle DEF at a point P . Suppose the ratio $PQ/PE = a/b$ where a and b are coprime positive integers. What is the remainder when ab is divided by 10^5 ?

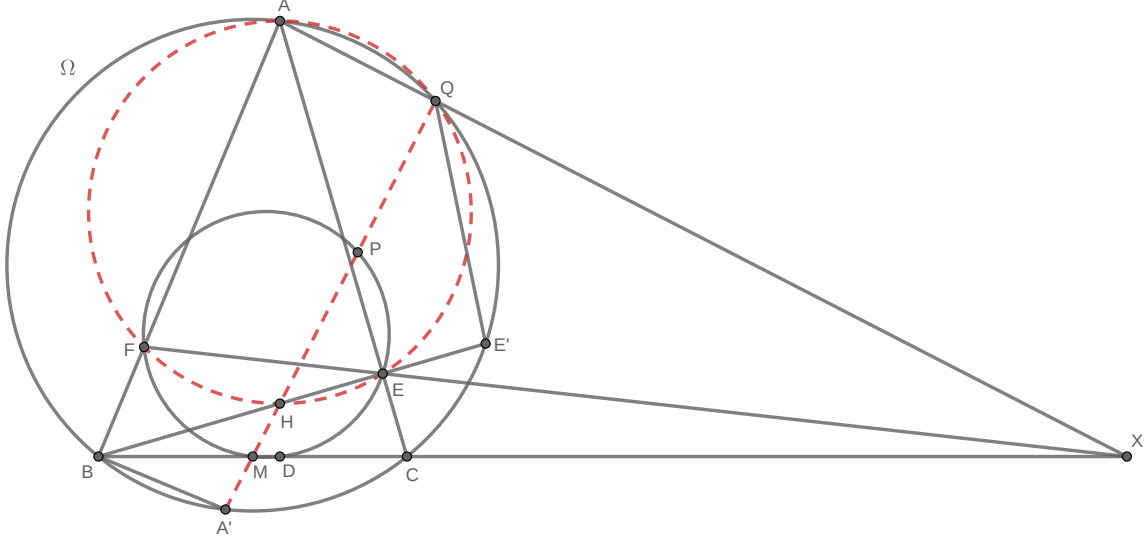
Answer: 11375

Remark: Relative to the AIMO3 version, we modified the final answer to ask for $a + b$ (rather than ab) to simplify the arithmetic.

Solution: Let M be the midpoint of BC , and let A' be the point diametrically opposite A on circle ABC . Since $AB = 26$ is the largest side and

$$26^2 < 25^2 + 17^2,$$

triangle ABC is acute. In particular, its orthocentre H lies inside triangle ABC .



Claim 1: Q, H, M , and A' are collinear.

Proof. Since BH and CA' are both perpendicular to AC , we have $BH \parallel CA'$. Similarly, $CH \parallel BA'$. Therefore $BHCA'$ is a parallelogram. Its diagonals BC and HA' bisect each other, so A', M , and H are collinear.

Now $BFEC$ is cyclic with diameter BC . Hence, by power of a point at X with respect to Ω and to circle $BFEC$,

$$XQ \cdot XA = XC \cdot XB = XE \cdot XF.$$

By the converse of power of a point, A, Q, E, F are cyclic.

Also, A, E, H, F are cyclic with diameter AH . Since the two circles through A, E, F coincide, A, Q, E, H, F are cyclic. Thus $\angle AQH = 90^\circ$. On the other hand, AA' is a diameter of Ω , so $\angle AQA' = 90^\circ$. Hence QH and QA' are the same line, and therefore Q, H, M , and A' are collinear. $\square$

Claim 2 (Nine-point Circle): The image of circle DEF under the homothety with centre H and scale factor 2 is Ω .

Proof. Let D' be the image of D under this homothety. Since H, D, A are collinear and $HD \perp BC$, the point D' is the reflection of H in the line BC . As H lies inside triangle ABC (since it is acute), we have

Let D' be the image of D under this homothety. Since H, D, A are collinear and $HD \perp BC$, the point D' is the reflection of H in the line BC . As ABC is acute, H lies on the same side of BC as

A , so D' lies on the opposite side of BC from A . We therefore have

$$\angle BD'C = \angle BHC = 180^\circ - A,$$

Hence $\angle BD'C + \angle BAC = 180^\circ$, and since A and D' are on opposite sides of BC , the points A, B, C, D' are cyclic. Thus D' lies on Ω .

Similarly, the images E' and F' of E and F lie on Ω . The image of circle DEF under the homothety is a circle through the three distinct points D', E', F' on Ω , so this image circle is exactly Ω . $\square$

Take E' as defined in Claim 2, so E' is the reflection of H in E . Since Q lies on Ω , its preimage under the same homothety is the midpoint of HQ ; this point lies on circle DEF . We now check that this is the required point P on segment HQ .

By Claim 1, the line HQ also contains A' , and the two intersections of this line with Ω are Q and A' . Under the homothety, their preimages on circle DEF are the midpoints of HQ and HA' , respectively. But M is the midpoint of HA' , since $BHCA'$ is a parallelogram. As H lies inside Ω , the points Q and A' lie on opposite rays from H , so M is not on segment HQ . Therefore the intersection of segment HQ with circle DEF is the midpoint of HQ , namely P . Hence

$$HP = PQ = \frac{HQ}{2}.$$

The homothety also sends segment PE to segment QE' , so $QE' = 2PE$. Hence

$$\frac{PQ}{PE} = \frac{HQ/2}{QE'/2} = \frac{QH}{QE'}.$$

Since B, E, H, E' are collinear and, by Claim 1, Q, H, A' are collinear,

$$\angle E'HQ = \angle BHA'.$$

Moreover, Q, E', A', B all lie on Ω , so

$$\angle QE'H = \angle QE'B = \angle QA'B = \angle HA'B.$$

Thus $\triangle QHE' \sim \triangle BHA'$ and so

$$\frac{QH}{QE'} = \frac{BH}{BA'}.$$

It remains only to evaluate BH/BA' .

First, in triangle ABA' , the side AA' is a diameter, so $\angle ABA' = 90^\circ$. Also $\angle BA'A = \angle BCA = C$, since both angles subtend chord BA . Hence $\angle BAA' = 90^\circ - C$, and the extended sine rule gives

$$BA' = 2R \cdot \sin(90^\circ - C) = 2R \cdot \cos C,$$

where R is the circumradius of ABC .

Similarly, in the right-angled triangles ADB and BDH , we have

$$BD = AB \cdot \cos B = 2R \cdot \sin C \cos B$$

and, since $BH \perp AC$, we have $\angle DBH = 90^\circ - C$. Therefore,

$$BH = \frac{BD}{\cos(90^\circ - C)} = \frac{2R \sin C \cos B}{\sin C} = 2R \cos B.$$

Putting everything together,

$$\frac{PQ}{PE} = \frac{BH}{BA'} = \frac{2R \cos B}{2R \cos C} = \frac{\cos B}{\cos C}.$$

Now set $x = BC$, $y = AC$, $z = AB$. By the cosine rule,

$$\cos B = \frac{x^2 + z^2 - y^2}{2xz} \quad \text{and} \quad \cos C = \frac{x^2 + y^2 - z^2}{2xy}.$$

Therefore

$$\frac{PQ}{PE} = \frac{y(x^2 + z^2 - y^2)}{z(x^2 + y^2 - z^2)}.$$

Substituting $x = 17$, $y = 25$, and $z = 26$ gives

$$\frac{PQ}{PE} = \frac{25(17^2 + 26^2 - 25^2)}{26(17^2 + 25^2 - 26^2)} = \frac{8500}{6188} = \frac{125}{91}.$$

Thus $a = 125$ and $b = 91$, so the required value is

$$a + b = \boxed{216}.$$

Markscheme: We follow a two-step process for scoring solutions.

Step 1. Decide whether the solution is essentially complete.

Step 2. If it is not essentially complete, award 1 mark for significant partial progress and 0 marks otherwise. If it is essentially complete, award 7 marks unless there is a minor error or omission, in which case award 6 marks.

Essentially complete solutions (6/7 marks)

A solution is *essentially complete* if it proves an expression for PQ/PE involving only basic properties of the triangle. Specifically, it must prove that

$$\frac{PQ}{PE} = \frac{S(R, x, y, z, \sin A, \sin B, \sin C, \cos A, \cos B, \cos C)}{T(R, x, y, z, \sin A, \sin B, \sin C, \cos A, \cos B, \cos C)}$$

where S and T are multivariate polynomials and the parameters are as defined in the solution above ($x = BC$, $y = AC$, and $z = AB$). For the avoidance of doubt, the expression **cannot** include other length terms or angles even if they can be shown to be equal to an expression involving terms above (e.g. BA' , $\sin \angle CA'B$).

A coordinate, vector, or trigonometric solution can also be essentially complete. To qualify, it must compute the relevant constructed objects exactly, not just numerically (i.e. quantities must be expressed exactly rather than as decimal approximations), and it must identify the correct intersection point P on segment HQ .

An essentially complete solution should receive 6 rather than 7 if it contains a minor error or omission: a gap that is not merely a small arithmetic slip, but which can be repaired without changing the main structure of the solution. Examples include:

- (ME1) Quoting one of the key geometric facts below without a sufficiently precise statement of what it says in this configuration (e.g. asserting Claim 2 without linking this to the nine-point circle configuration).
- (ME2) Reaching a final expression such as $\cos B / \cos C$ but omitting the side-length substitution or final reduction to the required ratio.

The following should *not* generally be penalised:

- (NME1) A single routine arithmetic slip after a correct exact expression for the ratio has been obtained.
- (NME2) Assuming the triangle is acute without proving this and/or using that the triangle is acute when chasing angles without explicitly stating this.
- (NME3) Failing to rule out the possibility that segment HQ intersects circle DEF at two points.
 - We feel that this and the point above are both sufficiently obvious from drawing a diagram (particularly given the side lengths are fixed here) and would not usually be penalised in an Olympiad competition.

The following results may be quoted, provided they are stated precisely:

- A' , M , and H collinear.
- $DEFM$ is cyclic as this is the nine-point circle of triangle ABC .
- Claim 2, provided this is related to the nine-point circle.
- Q lies on the line through A' , M , H —this must be related to Q being the A -queue point of triangle ABC .

Partial progress for incomplete solutions (0/1 marks)

Award 1 mark to an incomplete solution if it proves that either

$$\frac{PQ}{PE} = \frac{QH}{QE'} \quad \text{or} \quad \frac{PQ}{PE} = \frac{BH}{BA'}$$

where in the first case, E' is shown to be the second intersection of BH with Ω other than B (it may be initially defined differently).

For clarity, none of the following on its own is enough for the partial-progress mark:

- (NPP1) Proving only Claim 1, namely Q, H, A' collinear.
- (NPP2) Proving only Claim 2, namely the homothety/nine-point circle relation.
- (NPP3) Proving only preliminary cyclic facts such as $BFEC$ cyclic or $AEHF$ cyclic.
- (NPP4) Computing side-length data such as R , the area, $\cos B$, or $\cos C$ without connecting it to PQ/PE .
- (NPP5) Stating the final answer 216, or the ratio 125/91, without a derivation.
- (NPP6) Setting up coordinates or other computations without completing the proof.

As is the case at the IMO, computational solutions (e.g. Cartesian coordinates) will not generally be awarded partial progress marks unless the intermediate results are interpreted synthetically (i.e. in terms of the points and length ratios in the diagram).

Problem 3 (SINDEX)

Problem: For a finite, nonempty multiset S of positive integers strictly less than 20, we define its *index* to be the greatest positive integer n such that for every i with $1 \leq i \leq n$ we can partition S into multisets A_i and B_i in such a way that $A_i = i + B_i$. If no such n exists, we define the index to be equal to 0.

Let N denote the maximum index across all choices for S . Across all choices of S with index N , let M denote the smallest possible value for the maximum element in S .

Across all multisets S with index N and maximum element M , what is the smallest possible number of elements of S (counted with multiplicity)?

For two multisets X and Y and a positive integer l , we write $X = l + Y$ if $X = \{l + y : y \in Y\}$ where the right-hand side is interpreted as a multiset.

Original AIMO3 Formulation:

For a finite, nonempty multiset S of positive integers strictly less than 20, we define its *index* to be the greatest positive integer n such that for every i with $1 \leq i \leq n$ we can partition S into multisets A_i and B_i in such a way that $A_i = i + B_i$. If no such n exists, we define the index to be equal to 0.

Let N denote the maximum index across all choices for S . Across all choices of S with index N , let M denote the smallest possible value for the maximum element in S .

Across all multisets S with index N and maximum element M , what is the smallest possible number of elements of S (counted with multiplicity)?

For two multisets X and Y and a positive integer l , we write $X = l + Y$ if $X = \{l + y : y \in Y\}$ where the right-hand side is interpreted as a multiset.

Answer: 32

Solution: For a finite multiset S , define

$$P_S(x) = \sum_{s \in S} x^s.$$

We call a polynomial $Q(x)$ *set-like* if there exists a multiset T of positive integers such that $Q = P_T$. This is equivalent to saying that Q has non-negative integer coefficients and no constant term.

Claim 1: For a positive integer m , S has index at least m if and only if for each $1 \leq i \leq m$, $\frac{P_S(x)}{1+x^i}$ is a set-like polynomial.

Proof. Suppose first that, for a fixed positive integer i , we have a partition

$$S = A_i \sqcup B_i$$

with $A_i = i + B_i$. Then

$$P_{A_i}(x) = \sum_{a \in A_i} x^a = \sum_{b \in B_i} x^{b+i} = x^i P_{B_i}(x),$$

and hence

$$P_S(x) = P_{A_i}(x) + P_{B_i}(x) = (1 + x^i)P_{B_i}(x).$$

Therefore $P_S(x)/(1 + x^i) = P_{B_i}(x)$ is set-like.

Conversely, suppose that $P_S(x)/(1 + x^i)$ is set-like. Choose a multiset B_i such that

$$P_{B_i}(x) = \frac{P_S(x)}{1 + x^i},$$

and define $A_i = i + B_i$. Then

$$P_{A_i}(x) + P_{B_i}(x) = x^i P_{B_i}(x) + P_{B_i}(x) = P_S(x).$$

Since equality of these polynomials means equality of the multiplicities of every positive integer, A_i and B_i form the required partition of S . Applying this for every $1 \leq i \leq m$ proves the claim. $\square$

Throughout the remainder of the solution, we shall use the cyclotomic factorizations

$$\begin{aligned} 1 + x &= \Phi_2, & 1 + x^2 &= \Phi_4, & 1 + x^3 &= \Phi_2 \Phi_6, \\ 1 + x^4 &= \Phi_8, & 1 + x^5 &= \Phi_2 \Phi_{10}, & 1 + x^6 &= \Phi_4 \Phi_{12}, \\ 1 + x^7 &= \Phi_2 \Phi_{14}, \end{aligned}$$

where

$$\begin{aligned} \Phi_2 &= 1 + x, & \Phi_4 &= 1 + x^2, & \Phi_6 &= 1 - x + x^2, \\ \Phi_8 &= 1 + x^4, & \Phi_{10} &= 1 - x + x^2 - x^3 + x^4, \\ \Phi_{12} &= 1 - x^2 + x^4, & \Phi_{14} &= 1 - x + x^2 - x^3 + x^4 - x^5 + x^6. \end{aligned}$$

These distinct cyclotomic factors are well-known to be irreducible. Since every P_S also has zero constant term, any multiset with index at least m has P_S divisible by

$$x \operatorname{lcm}(1 + x, 1 + x^2, \dots, 1 + x^m).$$

We can expand the lcm as a product of the relevant cyclotomic factors.

Claim 2: $N \leq 6$.

Proof. If S had index at least 7, then by the above, $P_S(x)$ is divisible by

$$x \cdot \Phi_2 \cdot \Phi_4 \cdot \Phi_6 \cdot \Phi_8 \cdot \Phi_{10} \cdot \Phi_{12} \cdot \Phi_{14}.$$

P_S has degree at most 19 (since the degree is bounded by the largest element) but the above has degree

$$1 + 1 + 2 + 2 + 4 + 4 + 4 + 6 = 24$$

giving a contradiction. $\square$

Claim 3: $N \leq 5$.

Proof. Next suppose, for contradiction, that some S has index at least 6. Put

$$Q = x \cdot \Phi_2 \cdot \Phi_4 \cdot \Phi_6 \cdot \Phi_8 \cdot \Phi_{10} \cdot \Phi_{12}.$$

Then $Q \mid P_S$ by the discussion above and $\deg Q = 18$, so

$$P_S = (ax + b)Q$$

for some integers a, b , since Q is monic and P_S has integer coefficients. Since

$$Q(x) = x - x^2 + x^3 + \cdots \quad \text{and} \quad (ax + b)Q(x) = bx + (a - b)x^2 + (b - a)x^3 + \cdots,$$

nonnegativity of the coefficients of P_S at x^2 and x^3 gives $a = b$. From the x coefficient and S being nonempty, $a = b > 0$. But then

$$\frac{P_S}{1 + x} = aQ,$$

which is not set-like because Q has a negative coefficient. This contradicts the required condition for $i = 1$, proving the claim. $\square$

Claim 4: $N = 5$.

Proof. Consider

$$P_0(x) = x \prod_{j=1}^5 (1 + x^j).$$

This is set-like, has degree $1 + 1 + 2 + 3 + 4 + 5 = 16$, and satisfies

$$\frac{P_0(x)}{1 + x^i} = x \prod_{\substack{1 \leq j \leq 5 \\ j \neq i}} (1 + x^j), \quad i = 1, 2, 3, 4, 5,$$

which is set-like for each i . Thus a multiset of index at least 5 and maximum element 16 exists. Since $N \leq 5$, we have $N = 5$. $\square$

Claim 5: $M = 16$.

Proof. For contradiction, assume there exists a multiset with index 5 and maximum element less than 16. From the discussion above, we have that

$$R(x) = x \cdot \Phi_2 \cdot \Phi_4 \cdot \Phi_6 \cdot \Phi_8 \cdot \Phi_{10}$$

divides $P_S(x)$. We have

$$\deg R = 1 + 1 + 2 + 2 + 4 + 4 = 14$$

so, as in Claim 3, we must have that

$$P_S(x) = (ax + b)R(x)$$

for integers a and b . Also

$$R = x - x^2 + 2x^3 + \cdots - x^{13} + x^{14}.$$

The coefficients of x^2 and x^{14} in $(ax + b)R$ are respectively $a - b$ and $b - a$, so being set-like forces $a = b$. The coefficient of x is then a , so nonemptiness gives $a > 0$. Hence

$$\frac{P_S}{1+x} = aR,$$

which is not set-like because R has a negative coefficient.

As we have shown in Claim 4 that there exists a multiset with index 5 and maximum element 16, we are done. $\square$

Claim 6: For a multiset S attaining the bounds $N = 5$ and $M = 16$, the minimum possible number of elements is 32.

Proof. The number of elements of S is equal to $P_S(1)$. This shows the example given in Claim 4 has $2^5 = 32$ elements. It remains to show we cannot do better.

Since $R \mid P_S$ and $\deg R = 14$, we may write

$$P_S = (ax^2 + bx + c)R$$

for integers a, b, c , with $a \geq 1$ because the coefficient of x^{16} is positive as S must have largest element exactly 16. The condition for $i = 1$ says that

$$\frac{P_S(x)}{1+x} = (ax^2 + bx + c) \frac{R(x)}{1+x}$$

is set-like. Now

$$\frac{R(x)}{1+x} = x - 2x^2 + 4x^3 + \cdots - 2x^{12} + x^{13}.$$

Looking at the coefficients of x^{14} and x^3 in $P_S(x)/(1+x)$ gives

$$b - 2a \geq 0, \quad a - 2b + 4c \geq 0.$$

Thus $b \geq 2a \geq 2$, and

$$c \geq \frac{2b - a}{4} \geq \frac{3a}{4}.$$

Since a, c are integers and $a \geq 1$, this gives $c \geq 1$. Therefore

$$|S| = P_S(1) = (a + b + c)R(1) \geq (1 + 2 + 1)R(1) = 4 \cdot 8 = 32$$

where we have used $R(1) = 8$. This proves that 32 is indeed the minimum. □

Putting all these Claims together, we have shown that the final answer is 32.

Markscheme: We follow a two-step process for scoring solutions.

Step 1. Decide whether the solution is essentially complete.

Step 2. If it is not essentially complete, award 1 mark for significant partial progress and 0 marks otherwise. If it is essentially complete, award 7 marks unless there is a minor error or omission, in which case award 6 marks.

Essentially complete solutions (6/7 marks)

A solution is *essentially complete* if, allowing only minor slips of the kind described below, it proves the final answer 32. In particular, it must establish all three optimisation layers in the problem: the maximum possible index is $N = 5$, the minimum possible maximum element among multisets with index 5 is $M = 16$, and the minimum possible size among multisets with index 5 and maximum element 16 is 32.

The solution does not have to use the polynomial method above. However, any essentially complete solution must contain both upper-bound arguments and matching constructions for these three values. In a polynomial solution, this means that it is not enough to prove divisibility by the polynomials $1 + x^i$: wherever a construction is claimed, the relevant quotients must also be shown to be set-like, or the corresponding partitions must be given by another method.

An essentially complete solution should receive 6 rather than 7 if it contains a minor error or omission: a gap that is not merely a small arithmetic slip, but which can be repaired without changing the main structure of the solution. Examples include:

- (ME1) Using the correct least common multiple of the polynomials $1 + x^i$ but not explicitly justifying why the distinct factors may be multiplied together, for example by mentioning irreducibility, coprimality, or cyclotomic factors.
- (ME2) Correctly reducing the $N \leq 5$ argument to $P_S(x) = (ax + b)Q(x)$, but omitting a short justification that a and b are integers (if the solution requires this), or not explicitly pointing out the negative coefficient that makes $P_S(x)/(1 + x)$ fail to be set-like.
- (ME3) Giving the construction $P_S(x) = (1 + x)^2 R(x)$ but only sketching the verification that the five quotients $P_S(x)/(1 + x^i)$ are set-like. If the construction is presented as

$$P_S(x) = x \prod_{j=1}^5 (1 + x^j)$$

then this is immediate and can simply be acknowledged rather than requiring further justification.

- (ME4) Making a fixable error in the final coefficient inequalities used to prove $P_S(1) \geq 32$, while still setting up the correct representation $P_S(x) = (ax^2 + bx + c)R(x)$ and using the set-likeness of $P_S(x)/(1 + x)$.

The following should *not* generally be penalised:

- (NME1) Stating the standard factorisations into cyclotomic factors for $1 + x^i$, $1 \leq i \leq 7$, without proving the irreducibility of each factor, provided the factorisations and least common multiples are used correctly.
- (NME2) Omitting long polynomial expansions, provided the solution clearly states the necessary set-likeness checks and those checks are correct.

- (NME3) Minor arithmetic slips in displayed expansions that do not affect any sign check, degree bound, or final numerical value.
- (NME4) Not proving separately that the construction has index *exactly* 5, provided the solution has already proved the global upper bound $N \leq 5$ and has verified that the construction has index at least 5.
- (NME5) An isolated typo such as writing “set” instead of “multiset”, provided the solution otherwise treats multiplicities correctly.

The following are not essentially complete, even if they contain some correct calculations:

- (NEC1) A solution that proves $N \leq 6$ but does not rule out index 6.
- (NEC2) A solution that gives only upper bounds, or only a construction, but not both.
- (NEC3) A solution that assumes S is an ordinary set rather than a multiset, or otherwise imposes an extra condition not present in the problem.
- (NEC4) A polynomial solution that checks only divisibility by $1 + x^i$ in its construction and never addresses the non-negativity of the quotient coefficients.
- (NEC5) A solution that proves $N = 5$ but does not prove the two later minimisation statements, namely $M = 16$ and the final size bound 32.
- (NEC6) A solution that relies only on numerical search or pattern-spotting without a proof for the required upper bounds.

Partial progress for incomplete solutions (0/1 marks)

Award 1 mark to an incomplete solution if it contains a correct proof that $N \leq 6$ (Claim 2).

Do not award the partial-progress mark for any of the following on its own:

- (NPP1) Introducing the polynomial $P_S(x)$.
- (NPP2) Proving only the translation between partitions and the condition that $P_S(x)/(1+x^i)$ is set-like (Claim 1).
- (NPP3) Listing the factorisations of $1 + x^i$ without using them to obtain a valid degree contradiction or another meaningful bound.
- (NPP4) Guessing or displaying the final ‘optimal’ polynomial $P_S(x)$ without proving the necessary bounds.
- (NPP5) Numerical experimentation, checking a few examples, or making conjectures about the maximum possible index.

Problem 4 (COLOUR)

Problem: Linda and Sue play a game on a 2025×2025 grid of cells. Initially, every cell is grey. Linda moves first, and the players then alternate turns.

On each of Linda's turns, she chooses one cell and paints it lime, regardless of its current colour. On each of Sue's turns, she chooses an axis-aligned 2×2 block of cells containing at least one grey cell, and paints all four cells scarlet, regardless of their current colours.

After each move, if there are no grey cells left, the game ends immediately. Linda wants to maximise the number of cells whose final colour is lime, and Sue wants to maximise the number of cells whose final colour is scarlet.

- a) Prove that the game always ends after finitely many moves.
- b) Suppose both players play optimally. Determine the number of lime cells at the end of the game.

Original AIMO3 Formulation:

Linda and Sue play a game on a 2025×2025 grey square grid as follows. Linda starts by picking a cell of any colour and painting it lime. Then Sue picks a 2×2 square, at least one of which is grey, and paints all four cells scarlet (including any cells already painted lime). They alternate turns with the game finishing immediately once no cell is grey. Linda wants to maximise the number of lime cells and Sue wants to maximise the number of scarlet cells. If both players play optimally, the game will end with L lime cells. What is the remainder when L is divided by 10^5 ?

Answer: 26168

Remark: Relative to the AIMO3 version, we have made it more explicit how the game progresses, and that the players' objectives are determined by the final colours of the cells. We regard the intended meaning of the original formulation as clear in context, but these small changes remove possible sources of misreading. We have also added part *a)* to make explicit that the game always terminates, so that the optimisation question in part *b)* is well-defined.

Solution: We prove a slightly more general statement. For an $n \times n$ grid with $n \geq 2$, if both players play optimally then the final number of lime cells is

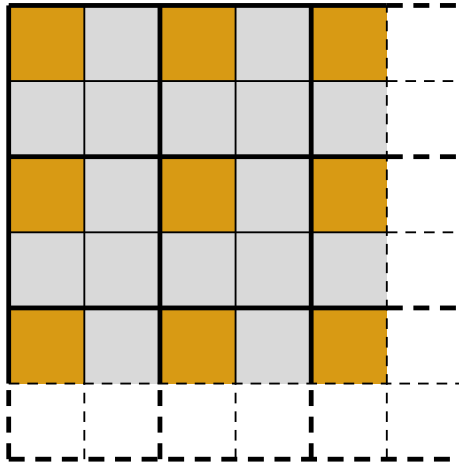
$$\left\lceil \frac{n}{2} \right\rceil^2 - 1.$$

Substituting $n = 2025$ gives

$$L = \left\lceil \frac{2025}{2} \right\rceil^2 - 1 = 1013^2 - 1 = \boxed{1026168}.$$

We first prove part *a*). On each of Sue's turns, she is required to choose a 2×2 square containing at least one grey cell. After she paints the square scarlet, at least one formerly grey cell is no longer grey. Linda's move cannot increase the number of grey cells. Therefore the number of grey cells strictly decreases on every Sue turn, and there are only finitely many grey cells to begin with. Hence the game must end after finitely many turns.

Partition the grid into a grid of 2×2 squares. If n is odd, as it is in the problem here, extend down and right to a $(n + 1) \times (n + 1)$ grid and partition there, then restrict to the original grid. We get $\left\lceil \frac{n}{2} \right\rceil^2$ regions which we call *compartments*, each of which has a unique top-left cell which we call *special*. Every 2×2 square of the grid, not necessarily a compartment, contains exactly one special cell.



Linda's strategy: Linda's strategy is to pick a non-special cell on her first move, then for every move after Sue's, to paint the special cell in Sue's previous 2×2 square lime. First, note that if Sue never paints a fixed special cell, then Linda never does either, so the game will not have ended. Thus, by the time the game has ended, Sue has painted all special cells at least once. Next, note that Linda will always paint over a non-grey cell after her first move, hence Sue makes the last move in order to remove the final grey cell(s).

Other than the unique special cell in this last 2×2 square, fix a special cell and consider the last time it is painted scarlet. Immediately after, it is painted lime and remains that way. Thus, other than the last special cell, all special cells are lime, and hence at least $\left\lceil \frac{n}{2} \right\rceil^2 - 1$ are lime at the end.

Sue's strategy: Sue's goal of maximising the number of scarlet cells is the same as minimising the number of lime cells since at the end of the game every cell is coloured one or the other. After Linda's first move, Sue chooses one of the four rotated compartment decompositions such that the lime cell does not lie in its own 1×1 compartment (which would be the bottom right cell for n odd). Then, Sue picks the compartment containing this lime cell, which necessarily contains a grey cell, and paints a 2×2 square which covers it (which may overlap a neighbouring compartment if the chosen compartment lies on the boundary). On Sue's subsequent turns, she selects any compartment with a grey cell and paints a legal 2×2 square covering that compartment.

After Sue picks a compartment, she will not pick it again as it does not contain any grey cells. Hence, the game ends on or before Sue's $\lceil \frac{n}{2} \rceil^2$ th move. Thus, there are at most as many lime cells remaining as there are Linda moves after Sue's first move, hence at most $\lceil \frac{n}{2} \rceil^2 - 1$ lime cells.

Putting these together gives the required value for L .

Markscheme: We follow a two-step process for scoring solutions.

Step 1. Decide whether the solution is essentially complete.

Step 2. If it is not essentially complete, award 1 mark for significant partial progress and 0 marks otherwise. If it is essentially complete, award 7 marks unless there is a minor error or omission, in which case award 6 marks.

Essentially complete solutions (6/7 marks)

Let $M = 1013^2 - 1$, the correct value of L .

A solution is *essentially complete* if it proves that the game always terminates and proves that $L \in \{M - 1, M, M + 1\}$. This requires providing and proving a strategy for Linda to obtain at least $M - 1$ lime cells and a strategy for Sue to restrict Linda to at most $M + 1$ lime cells.

An essentially complete solution that fails to prove the exact value $L = M$ should score 6 marks.

An essentially complete solution should also receive 6 rather than 7 if it contains a minor error or omission: a gap which is not merely a small arithmetic slip, but which can be repaired without changing the main structure of the solution. Examples include:

- (ME1) In Linda's strategy, failing to explain why all special cells except possibly the one in Sue's final square are lime at the end.
- (ME2) In Sue's strategy, failing to justify that boundary compartments can be covered by legal 2×2 squares.
- (ME3) In Sue's strategy, failing to explain why Linda's first lime cell is overwritten on Sue's first move.
- (ME4) Making a fixable off-by-one error in counting Sue's compartment-clearing moves or Linda's remaining moves, while the intended strategy and final value are otherwise clear.

We generally will not penalise minor arithmetic mistakes unless they are integral to the solution. The final answer may be left as a basic arithmetic expression, such as $1013^2 - 1$; it does not have to be explicitly calculated as 1026168 to score full marks.

Partial progress for incomplete solutions (0/1 marks)

If a solution is not essentially complete, it should score 1 if it proves at least one of the following substantial one-sided results.

- (PP1) A valid Linda strategy guaranteeing at least

$$\left\lfloor \frac{2025}{2} \right\rfloor^2 - 1 = 1012^2 - 1 = 1024143$$

lime cells at the end. For example, this might be a correct special-cell strategy on a 2024×2024 sub-board, even if it does not handle the last row and column sharply.

- (PP2) A valid Sue strategy guaranteeing at most

$$\left\lceil \frac{2025}{2} \right\rceil^2 = 1013^2 = 1026169$$

lime cells at the end. For example, this might be a compartment-clearing strategy with a remaining unresolved off-by-one.

The following do *not* count as significant partial progress on their own.

- (NPP1) Proving only that the game terminates.
- (NPP2) Stating the correct answer, or the correct expression $1013^2 - 1$, without a proof of optimality.
- (NPP3) Checking small cases or giving numerical evidence without a strategy for the 2025×2025 board.
- (NPP4) Giving only trivial bounds, such as $0 \leq L \leq 2025^2$.
- (NPP5) Describing a 2×2 tiling of the board without deriving a substantial lower or upper bound from it.
- (NPP6) Giving a strategy that works only if the opponent cooperates, rather than against every legal move of the opponent.

Problem 5 (POLYGO)

Problem: A polynomial $P(x)$ is called *good* if it has integer coefficients and:

- It is a monic polynomial of degree 7;
- The absolute values of all of the coefficients of P are odd and strictly less than 2025^2 ;
- The absolute values of all the coefficients of P are roots of P .

How many good polynomials are there?

Original AIMO3 Formulation:

A polynomial $P(x)$ is called *good* if it has integer coefficients and:

- It is a monic polynomial of degree 7;
- The absolute values of all of the coefficients of P are odd and strictly less than 2025^2 ;
- The absolute values of all the coefficients of P are roots of P .

There are N good polynomials. What is the remainder when N is divided by 10^5 ?

Answer: 50968

Remark: Relative to the AIMO3 version, we removed the modulo 10^5 answer modification as we are not restricted to 5-digit answers in the Proof Pilot.

Solution: Write

$$P(x) = x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0,$$

and put $M = 2025^2$. Let N be the largest absolute value of any coefficient of P , including the leading coefficient. Then N is odd, $1 \leq N < M$, and both 1 and N are roots of P .

First suppose $N = 1$. Then each of $a_0, a_1, \dots, a_6$ is equal to 1 or -1 . Since 1 is a root,

$$0 = P(1) = 1 + a_6 + a_5 + \dots + a_0.$$

Thus exactly three of $a_0, a_1, \dots, a_6$ are equal to 1, and the remaining four are equal to -1 . This gives

$$\binom{7}{3} = 35$$

polynomials.

We present two approaches for classifying the cases where $N \geq 3$.

Approach 1: Right to Left

We determine the coefficients in pairs, starting from the constant term. We shall repeatedly use the following simple observation: if b is odd, $|b| \leq N$, $\varepsilon \in \{\pm 1\}$, and

$$b + \varepsilon \equiv 0 \pmod{N},$$

then $b = -\varepsilon$. Indeed, $b + \varepsilon$ is even and has absolute value at most $N + 1 < 2N$, while the only multiples of the odd number N in this range are $-N, 0, N$, of which only 0 is even.

Since $P(N) = 0$, reducing modulo N gives $a_0 \equiv 0 \pmod{N}$. As $0 < |a_0| \leq N$ and is odd, we have $a_0 = \varepsilon_0 N$ for some $\varepsilon_0 \in \{\pm 1\}$. Dividing $P(N) = 0$ by N and reducing modulo N gives

$$a_1 + \varepsilon_0 \equiv 0 \pmod{N},$$

so by the observation $a_1 = -\varepsilon_0$. Hence

$$a_1 \in \{\pm 1\}, \quad a_0 = -a_1 N.$$

The same argument may now be repeated after deleting this cancelling pair. Since $a_1 N + a_0 = 0$, dividing the remaining equation by N^2 gives

$$0 = N^5 + a_6 N^4 + a_5 N^3 + a_4 N^2 + a_3 N + a_2.$$

Applying the same two congruence steps gives

$$a_3 \in \{\pm 1\}, \quad a_2 = -a_3 N.$$

Deleting this pair and dividing by N^4 leaves

$$0 = N^3 + a_6 N^2 + a_5 N + a_4,$$

so similarly

$$a_5 \in \{\pm 1\}, \quad a_4 = -a_5 N.$$

Finally the remaining part of $P(N) = 0$ is $N^7 + a_6N^6 = 0$, so $a_6 = -N$.

Therefore every good polynomial with $N \geq 3$ has the form

$$P(x) = x^7 - Nx^6 + s_1(x^5 - Nx^4) + s_2(x^3 - Nx^2) + s_3(x - N),$$

where $s_1, s_2, s_3 \in \{\pm 1\}$. Since 1 is a root,

$$0 = P(1) = (1 - N)(1 + s_1 + s_2 + s_3).$$

As $N > 1$, this is equivalent to

$$s_1 + s_2 + s_3 = -1.$$

Thus exactly one of s_1, s_2, s_3 is 1 and the other two are -1 . There are 3 ways to assign these values.

Conversely, for every odd integer N with $3 \leq N < M$, every polynomial of the displayed form with exactly one of s_1, s_2, s_3 equal to 1 is good: its coefficient absolute values are just 1 and N , the equation $P(N) = 0$ follows from the cancellation in each block, and $P(1) = 0$ follows from $1 + s_1 + s_2 + s_3 = 0$.

It remains to count the possible values of N . Since the coefficient absolute values must be strictly less than $M = 2025^2$, and M is odd, the odd values with $N \geq 3$ are

$$3, 5, 7, \dots, M - 2,$$

so there are $(M - 3)/2$ choices.

Approach 2: Left to Right

We begin with the observation that for any positive integer t

$$N^t + N^{t-1} + \dots + N + 1 = N^t + \frac{N^t - 1}{N - 1} < 2N^t. \quad (*)$$

We also note that, from the third condition, $P(N) = 0$. If $a_6 \geq 2 - N$ then

$$\begin{aligned} 0 = P(N) &= N^7 + a_6N^6 + a_5N^5 + \dots + a_1N + a_0 \\ &\geq N^7 + (2 - N)N^6 - N \cdot N^5 - \dots - N \cdot N - N \\ &= 2N^6 - N \cdot \underbrace{(N^5 + N^4 + \dots + 1)}_{< 2N^5 \text{ by } (*)} \\ &> 2N^6 - N(2N^5) = 0 \end{aligned}$$

which is a contradiction. Thus, a_6 is equal to either $1 - N$ or $-N$. As N and a_6 are both odd, it follows that $a_6 = -N$.

Now we have

$$0 = P(N) = N^7 - N \cdot N^6 + a_5N^5 + \dots + a_1N + a_0 = a_5N^5 + \dots + a_1N + a_0.$$

Subtracting a_5N^5 from both sides and applying the triangle inequality

$$\begin{aligned} |-a_5N^5| &= |a_5|N^5 = |a_4N^4 + \dots + a_1N + a_0| \\ &\leq |a_4|N^4 + \dots + |a_1|N + |a_0| \\ &\leq N \cdot N^4 + \dots + N \cdot N + N \\ &< N(2N^4) = 2N^5. \end{aligned} \quad (\text{By } (*))$$

Thus, $|a_5| < 2$ which means, since a_5 is odd, that $a_5 = \pm 1$. Noting that

$$0 = P(N) = a_5 N^5 + a_4 N^4 + \cdots + a_1 N + a_0 = \pm (N^5 \pm a_4 N^4 + \cdots \pm a_1 N \pm a_0)$$

we can repeat the argument from above (for $a_6 = -N$) to get $\pm a_4 = -N$ so $a_4 = -a_5 N$. We then have

$$0 = P(N) = a_5 N^5 - (a_5 N) N^4 + a_3 N^3 + \cdots + a_1 N + a_0 = a_3 N^3 + \cdots + a_1 N + a_0.$$

Again, we can repeat the argument that gave $a_5 = \pm 1$ and $a_4 = -a_5 N$ to get $a_3 = \pm 1$ and $a_2 = -a_3 N$.

Lastly, we note that $0 = P(N) = a_1 N + a_0$ so, by the same argument, $a_1 = \pm 1$ and $a_0 = -a_1 N$. Putting these all together we have

$$P(x) = x^7 - Nx^6 \pm (x^5 - Nx^4) \pm (x^3 - Nx^2) \pm (x - N).$$

The count is then the same as the first approach.

Combining with the 35 choices for P when $N = 1$, we get

$$35 + 3 \cdot \frac{M - 3}{2} = \frac{3M + 61}{2}$$

choices for P in total. Substituting in $M = 2025^2$, we get the answer

$$\frac{3 \cdot 2025^2 + 61}{2} = \boxed{6150968}.$$

Markscheme: We follow a two-step process for scoring solutions.

Step 1. Decide whether the solution is essentially complete.

Step 2. If it is not essentially complete, award 1 mark for significant partial progress and 0 marks otherwise. If it is essentially complete, award 7 marks unless there is a minor error or omission, in which case award 6 marks.

Essentially complete solutions (6/7 marks)

A solution is *essentially complete* if, for $N \geq 3$, it shows that any good P has the form

$$P(x) = x^7 - Nx^6 \pm (x^5 - Nx^4) \pm (x^3 - Nx^2) \pm (x - N)$$

where N is the largest absolute value of a coefficient.

Any solution that fails to then reduce this to the correct count of good polynomials will score 6.

We will not penalise arithmetic errors in computing the final answer (e.g. if there is a correct proof that there are $\frac{3M+61}{2}$ good polynomials but the calculation goes wrong when substituting in 2025^2).

Other examples of minor errors or omissions that would result in a score of 6 include:

- (ME1) Using $N \leq 2025^2$ instead of $N < 2025^2$ in the final count, thereby allowing the inadmissible value $N = 2025^2$.
- (ME2) Counting the odd values of $N \geq 3$ as $(M-1)/2$ rather than $(M-3)/2$, where $M = 2025^2$.
- (ME3) Omitting the $N = 1$ case, or counting it in a way that is not a minor arithmetic slip.
- (ME4) Failing to impose $P(1) = 0$ on the three signs, for example giving 8 sign choices instead of 3.
- (ME5) Not checking the converse direction when the construction is not already visibly reversible.

The following should *not* generally be penalised:

- (NME1) A simple arithmetic slip in evaluating 2025^2 or the final division, provided the correct expression $35 + 3(M-3)/2$ or $(3M+61)/2$ has been obtained.
- (NME2) Not explicitly stating that $M = 2025^2$ is odd, if the subsequent range $3, 5, \dots, M-2$ is counted correctly.
- (NME3) A very brief converse check, provided it clearly notes that the only coefficient absolute values in the constructed polynomial are 1 and N and that both are roots.

Partial progress for incomplete solutions (0/1 marks)

An incomplete solution scores 1 mark if it makes a correct, non-trivial structural deduction in the case $N \geq 3$. Any one of the following is sufficient:

- (PP1) Proving that $a_6 = -N$.
- (PP2) Proving that $a_0 = \pm N$ **and** $a_1 = \mp 1$ (where the sign choices are opposite).
- (PP3) Proving any full adjacent pair relation, such as $a_4 = -a_5N$ with $a_5 \in \{\pm 1\}$.

The following do *not* by themselves constitute partial progress:

- (NPP1) Counting the $N = 1$ case only.
- (NPP2) Merely observing that $P(1) = 0$ or $P(N) = 0$.
- (NPP3) Making claims that $a_6 = -N$ or $a_0 = \pm N$ without proof.
- (NPP4) Finding the correct numerical answer by search, guessing, or pattern-spotting without a proof of the main structure.
- (NPP5) Counting sign patterns for the block form when the block form has not been justified.

Problem 6 (AMAZIN)

Problem: Let ABC be an acute triangle with $\angle CBA = 75^\circ$, orthocentre H , and circumcircle Ω . Let line AH meet Ω again at $H_A \neq A$. Similarly, let line CH meet Ω again at H_C . Let the circle centred at B with radius BH meet the line BC at X on the opposite side of B from C . Assume that lines XH_A and CH are not parallel, and let them intersect at P . Let lines XH_C and AH intersect at Q .

Let n be an odd positive integer. We call a triangle ABC *n-good* if $\frac{CP}{CH} = \sqrt{n}$ and the length AQ is a positive integer. We say n is *amazing* if there exists an *n-good* triangle where the length H_CQ is also a positive integer.

Let $1 \leq n_1 < n_2 < n_3 < \dots$ be a list of all amazing, odd positive integers.

Let x be the largest real number for which there exists a positive real number y such that for all sufficiently large i , $n_{2i} \geq y \cdot x^i$. For this choice of x , let z be the largest real number such that for all sufficiently large i , $n_{2i} \geq z \cdot x^i$.

It is possible to write $x + 2016z = p + q\sqrt{r}$ where p , q , and r are positive integers with r square-free. What is $p + q + r$?

Original AIMO3 Formulation:

Let ABC be an acute triangle with $\angle CBA = 75^\circ$, orthocentre H , and circumcircle Ω . Let AH meet BC at D and Ω again at H_A . Similarly, let CH meet Ω again at H_C . Let the circle centred at B with radius BH meet BC at X on the opposite side of B as C . Let lines XH_A and CH intersect at P and lines XH_C and AH intersect at Q .

Let n be an odd positive integer. We call a triangle ABC *n-good* if $\frac{CP}{CH} = \sqrt{n}$ and the length AQ is a positive integer. We say n is *amazing* if there exists an *n-good* triangle where the length H_CQ is also a positive integer.

Let $1 \leq n_1 < n_2 < n_3 < \dots$ be a list of all amazing, odd positive integers.

Let x be the largest real number for which there exists a positive real number y such that for all sufficiently large i , $n_{2i} \geq y \cdot x^i$. For this choice of x , let z be the largest real number such that for all sufficiently large i , $n_{2i} \geq z \cdot x^i$.

It is possible to write $x + 2016z = p + q\sqrt{r}$ where p , q , and r are positive integers with r square-free. What is the remainder when $p + q + r$ is divided by 10^5 ?

Answer: 31070

Remark: Relative to the AIMO3 version, we have made explicit the assumption that XH_A and CH are not parallel, so that the point P is well defined. This does not affect the answer, since in the parallel case P is undefined and the triangle is not *n-good*. We have also inserted the word ‘line’ in a few places to avoid any misunderstandings that these referred to segments. The point D was removed from the problem statement as it was not actually used. Finally, we removed the modulo 10^5 answer modification since it did not actually have any effect on the final answer here.

Solution: We use directed angles modulo 180° . This is especially important here because there are two possible configurations, depending on the relative positions of P and Q .

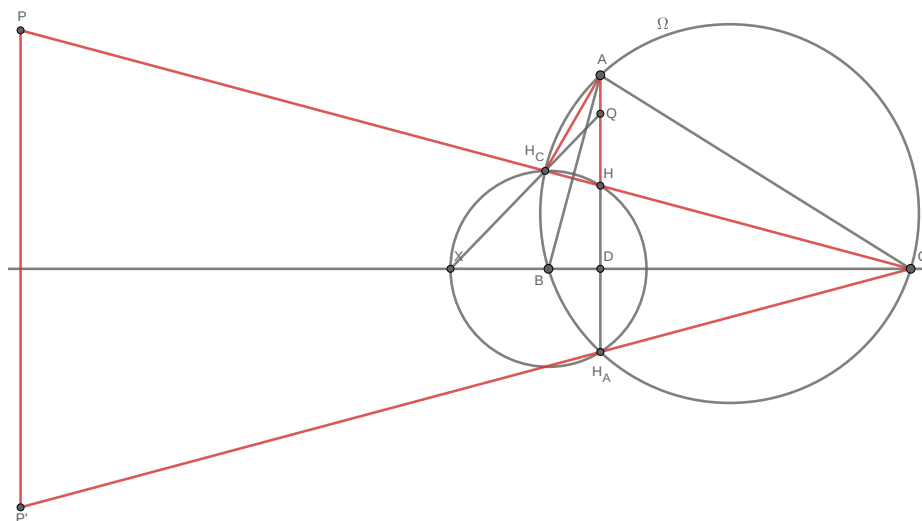


Figure 1: The *inside* configuration

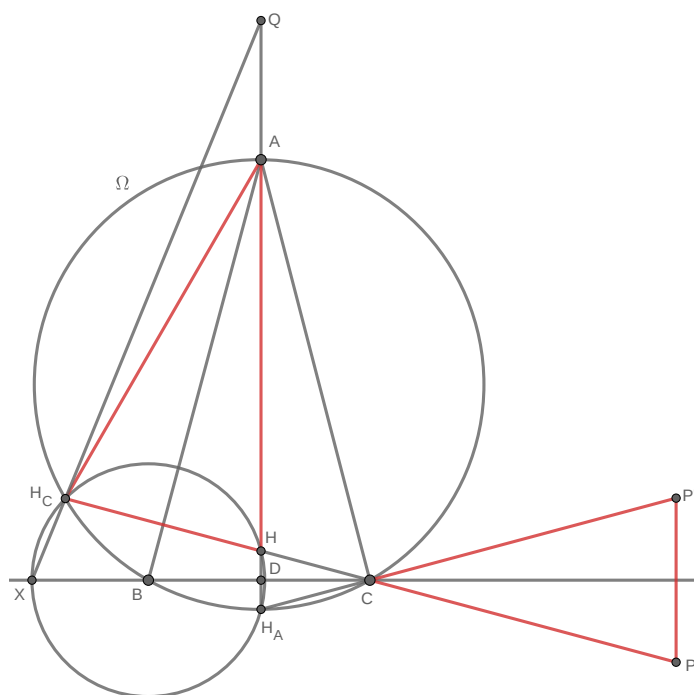


Figure 2: The *outside* configuration

We will determine exactly when each configuration occurs after proving the following geometric relation.

Claim: We have

$$\frac{CP}{CH} = \frac{AH}{AQ},$$

where the length ratios are directed.

Proof. Let P' be the reflection of P in BC . It's well-known that H_A and H_C are respectively the reflections of H in BC and AB , so

$$CP = CP', \quad AH = AH_C,$$

and also $P' \in CH_A$ and $PP' \parallel HH_A$.

Also $BH = BX = BH_A = BH_C$, so X, H_A, H, H_C are concyclic with centre B . Now

$$\angle HH_C A = \angle CH_C A = \angle CH_A H = \angle CP' P.$$

Therefore the isosceles triangles AHH_C and PCP' are similar, with correspondence

$$A \leftrightarrow C, \quad H \leftrightarrow P, \quad H_C \leftrightarrow P'.$$

It remains to identify the point on CP corresponding to $Q \in AH$. Using the reflection in BC and the circle through X, H_A, H, H_C , we get

$$\angle HP' P = \angle P' P H_A = \angle HH_A X = \angle HH_C X = \angle HH_C Q.$$

Thus, under the above similarity, the line $H_C Q$ corresponds to the line $P' H$. Hence Q corresponds to H , since Q lies on AH and H lies on CP . Therefore AQ corresponds to CH , while AH corresponds to CP , giving

$$\frac{CP}{CH} = \frac{AH}{AQ},$$

as required. □

We next seek to understand what the possible values for $\frac{CP}{CH}$ are and when we get the inside versus outside configurations.

Set $\gamma = \angle ACB$. Since $\angle CBA = 75^\circ$ and ABC is acute, we have $15^\circ < \gamma < 90^\circ$.

Choose Cartesian coordinates with $A = (0, 0)$ and AH on the positive x -axis. Since

$$\angle BAH = 90^\circ - \angle CBA = 15^\circ,$$

we may take AB to make angle -15° with the x -axis. Let $D = (d, 0)$ be where AH intersects BC and set

$$t = \tan 15^\circ = 2 - \sqrt{3}.$$

Then

$$B = (d, -dt), \quad C = (d, d \cot \gamma),$$

and the orthocentre is

$$H = (h, 0), \quad h = d(1 - t \cot \gamma).$$

Moreover

$$BH = dt \csc \gamma, \quad X = (d, -dt(1 + \csc \gamma)),$$

while the reflection of H in AB is

$$H_C = (h \cos 30^\circ, -h \sin 30^\circ).$$

Let $\lambda = \frac{AQ}{AH}$ be the directed ratio, positive on the ray AH and negative on the opposite ray. Intersecting XH_C with the x -axis gives

$$\lambda = \frac{t\sqrt{3}(\sin \gamma + 1) - \sin \gamma}{2t(\sin \gamma + 1) - \sin \gamma + t \cos \gamma}.$$

To see that λ is strictly decreasing, set

$$s = \frac{1 + \sin \gamma}{\cos \gamma} = \tan \left(45^\circ + \frac{\gamma}{2} \right).$$

As γ increases from 15° to 90° , the quantity $45^\circ + \frac{\gamma}{2}$ increases from 52.5° to 90° , so s is strictly increasing. Using

$$\sin \gamma = \frac{s^2 - 1}{s^2 + 1}, \quad \cos \gamma = \frac{2s}{s^2 + 1}, \quad 1 + \sin \gamma = \frac{2s^2}{s^2 + 1},$$

and also

$$t\sqrt{3} = 1 - 2t, \quad t^2 = 4t - 1,$$

the expression for λ simplifies to

$$\begin{aligned} \lambda &= \frac{t\sqrt{3}(1 + \sin \gamma) - \sin \gamma}{2t(1 + \sin \gamma) - \sin \gamma + t \cos \gamma} \\ &= \frac{1 - t^2 s^2}{1 + 2ts + t^2 s^2} = \frac{1 - ts}{1 + ts} = -1 + \frac{2}{1 + ts}. \end{aligned}$$

Since s is strictly increasing and t is fixed, this shows that λ is a decreasing function of γ . The endpoint values are

$$\lim_{\gamma \downarrow 15^\circ} \lambda = \frac{2}{1 + \sqrt{2} + \sqrt{3}}, \quad \lambda(60^\circ) = 0, \quad \lim_{\gamma \uparrow 90^\circ} \lambda = -1$$

and by continuity, λ takes each value between these points as γ varies.

By the Claim, using ordinary lengths in the definition of n -good,

$$\sqrt{n} = \frac{CP}{CH} = \frac{AH}{AQ} = \frac{1}{|\lambda|}.$$

Hence

$$\begin{aligned} 15^\circ < \gamma < 60^\circ : & \text{ inside configuration, } \frac{CP}{CH} \in \left(\frac{1 + \sqrt{2} + \sqrt{3}}{2}, \infty \right), \\ 60^\circ < \gamma < 90^\circ : & \text{ outside configuration, } \frac{CP}{CH} \in (1, \infty). \end{aligned}$$

Since

$$\sqrt{3} < \frac{1 + \sqrt{2} + \sqrt{3}}{2} < \sqrt{5},$$

the first case can occur for odd n exactly when $n \geq 5$, while the second can occur for odd n exactly when $n \geq 3$.

Now suppose ABC is n -good, and write $AQ = q$, where q is a positive integer. The Claim gives

$$AH = q\sqrt{n}.$$

Also $AH_C = AH$, and, since H_C is the reflection of H in AB ,

$$\angle H_CAH = 2\angle BAH = 30^\circ.$$

If Q lies on the ray AH , the cosine rule in triangle AH_CQ gives

$$H_CQ^2 = q^2(n + 1 - \sqrt{3n}).$$

If Q lies on the opposite ray, the angle at A is 150° , and

$$H_CQ^2 = q^2(n + 1 + \sqrt{3n}).$$

Thus H_CQ can be integral only if $n + 1 \pm \sqrt{3n}$ is a rational square. Hence $\sqrt{3n}$ is rational, so, since n is an integer, $3n$ is a square. Writing $3n = (3m)^2$, we get

$$n = 3m^2,$$

where m is an odd positive integer. Then $n + 1 \pm \sqrt{3n} = 3m^2 + 1 \pm 3m$ is an integer; if it is a rational square, it must in fact be an integer square.

Conversely, if one of the corresponding integers below is a square, then the range analysis above lets us choose the required shape of triangle, and then scale it so that $AQ = 1$. This gives an n -good triangle for which H_CQ is an integer. Therefore the amazing integers are exactly those obtained from one of

$$\begin{cases} u^2 = 3m^2 - 3m + 1, & m \geq 3 \text{ odd,} & \text{inside configuration,} \\ v^2 = 3m^2 + 3m + 1, & m \geq 1 \text{ odd,} & \text{outside configuration.} \end{cases}$$

The first equation is equivalent to

$$(2u)^2 - 3(2m - 1)^2 = 1,$$

with $2m - 1 \equiv 1 \pmod{4}$ and $2m - 1 \geq 5$. The second equation is equivalent to

$$(2v)^2 - 3(2m + 1)^2 = 1,$$

with $2m + 1 \equiv 3 \pmod{4}$ and $2m + 1 \geq 3$.

Both are instances of the Pell equation

$$N^2 - 3M^2 = 1.$$

By standard theory of Pell's equation, since the fundamental solution is $(N_0, M_0) = (2, 1)$, all its positive solutions are given by

$$N_k + M_k\sqrt{3} = (2 + \sqrt{3})^{k+1}, \quad k \geq 0.$$

Equivalently,

$$N_{k+1} = 2N_k + 3M_k, \quad M_{k+1} = N_k + 2M_k,$$

starting from $(N_0, M_0) = (2, 1)$.

We only need solutions with N even. Reducing the recurrence modulo 2, we see that N_k is even exactly when k is even. Write

$$S_j = M_{2j} \quad (j \geq 0).$$

The two-step recurrence gives

$$M_{k+2} = 4N_k + 7M_k,$$

so

$$S_{j+1} \equiv -S_j \pmod{4}.$$

Since $S_0 = 1$, it follows that

$$S_j \equiv (-1)^j \pmod{4}.$$

Thus the inside solutions, for which $M = 2m - 1 \equiv 1 \pmod{4}$, correspond to even j , while the outside solutions, for which $M = 2m + 1 \equiv 3 \pmod{4}$, correspond to odd j . The only even- j solution excluded by the condition $m \geq 3$ is $j = 0$, since $S_0 = 1 = 2m - 1$ gives $m = 1$.

Consequently, after removing this excluded case, the amazing integers occur in increasing order as

$$\sqrt{n_j/3} = \frac{S_j + (-1)^j}{2}, \quad j \geq 1.$$

Indeed, the sequence S_j is strictly increasing, so these values of m are strictly increasing.

Define

$$\alpha = (2 + \sqrt{3})^2 = 7 + 4\sqrt{3}, \quad \beta = (2 - \sqrt{3})^2 = 7 - 4\sqrt{3} = \alpha^{-1}.$$

Then $0 < \beta < 1$, and

$$S_j = M_{2j} = \frac{(2 + \sqrt{3})\alpha^j - (2 - \sqrt{3})\beta^j}{2\sqrt{3}}.$$

For the even-indexed amazing numbers,

$$n_{2i} = 3 \left(\frac{S_{2i} + 1}{2} \right)^2.$$

Hence

$$\frac{n_{2i}}{\alpha^{4i}} = \frac{3}{4} \left(\frac{S_{2i} + 1}{\alpha^{2i}} \right)^2 \xrightarrow{i \rightarrow \infty} \frac{3}{4} \left(\frac{2 + \sqrt{3}}{2\sqrt{3}} \right)^2 = \frac{7 + 4\sqrt{3}}{16} = \frac{\alpha}{16}.$$

Moreover,

$$S_{2i} + 1 = \frac{2 + \sqrt{3}}{2\sqrt{3}} \alpha^{2i} + 1 - \frac{2 - \sqrt{3}}{2\sqrt{3}} \beta^{2i} > \frac{2 + \sqrt{3}}{2\sqrt{3}} \alpha^{2i},$$

because $0 < \frac{2 - \sqrt{3}}{2\sqrt{3}} \beta^{2i} < 1$. Therefore

$$n_{2i} > \frac{\alpha}{16} \alpha^{4i}$$

for all $i \geq 1$. From the definitions of x and z , we obtain

$$x = \alpha^4, \quad z = \frac{\alpha}{16}.$$

Finally,

$$\begin{aligned}x + 2016z &= (7 + 4\sqrt{3})^4 + 126(7 + 4\sqrt{3}) \\ &= 19699 + 11368\sqrt{3}.\end{aligned}$$

Therefore

$$p + q + r = 19699 + 11368 + 3 = \boxed{31070}.$$

Markscheme: We follow a two-step process for scoring solutions.

Step 1. Decide whether the solution is essentially complete.

Step 2. If it is not essentially complete, award 1 mark for significant partial progress and 0 marks otherwise. If it is essentially complete, award 7 marks unless there is a minor error or omission, in which case award 6 marks.

Essentially complete solutions (6/7 marks)

A solution is *essentially complete* if, allowing only minor errors or omissions of the kind described below, it gives a coherent proof of all of the following:

(EC1) A valid geometric reduction, including the directed-ratio relation

$$\frac{CP}{CH} = \frac{AH}{AQ},$$

or an equivalent relation strong enough to connect the condition $\frac{CP}{CH} = \sqrt{n}$ with the position of Q on the line AH .

(EC2) A correct analysis of the two configurations, including the possible ranges of $\frac{CP}{CH}$ in the inside and outside cases, or an equivalent existence argument that justifies exactly which odd values of n can occur in each case.

(EC3) A correct integrality reduction showing that every amazing integer has the form $n = 3m^2$ with m odd, and that the remaining conditions are exactly

$$\begin{cases} u^2 = 3m^2 - 3m + 1, & m \geq 3 \text{ odd}, \\ v^2 = 3m^2 + 3m + 1, & m \geq 1 \text{ odd}. \end{cases}$$

This should include the converse direction, for example by explaining that the triangle can be scaled so that $AQ = 1$.

(EC4) A correct solution of the resulting Pell equation, with the parity and congruence restrictions handled correctly, leading to

$$\sqrt{n_j/3} = \frac{S_j + (-1)^j}{2}, \quad j \geq 1,$$

where $S_j = M_{2j}$ and $N_k^2 - 3M_k^2 = 1$.

(EC5) A correct asymptotic evaluation of the even-indexed amazing integers, giving

$$x = (7 + 4\sqrt{3})^4, \quad z = \frac{7 + 4\sqrt{3}}{16},$$

and hence the final value 31070.

An essentially complete solution should receive 6 rather than 7 if it contains a minor error or omission: a gap that is not merely a small arithmetic slip, but which can be repaired without changing the main structure of the solution. Examples include:

- (ME1) Deriving the formula for λ but giving an incomplete justification for the range of possible (signed) values it can take. For example, calculating the correct endpoint values, but giving an incorrect proof of monotonicity, such as saying only that the numerator and denominator are decreasing.
- (ME2) Failing to justify why different (integer) choices of AQ do not lead to more possibilities for amazing n .
- (ME3) Computing the limiting constant for z correctly but omitting the short argument that the inequality needed in the definition of z holds eventually, not just that z is the limiting value.

The following should *not* generally be penalised:

- (NME1) Routine arithmetic or simplification slips that do not affect the final structural conclusions, such as a harmless intermediate trigonometric simplification that is corrected later.
- (NME2) Invoking standard Pell equation theory without reproving it from first principles, provided the relevant parity, congruence, and indexing restrictions are dealt with correctly.
- (NME3) Not separately proving that the listed amazing integers are already in increasing order, provided this is clear from the monotonicity of the Pell parameter used.
- (NME4) A simple final addition slip in $19699 + 11368 + 3$, provided the values of p , q , and r have been identified correctly.

The following are *not* essentially complete, even if they contain some substantial correct work:

- (NEC1) A solution that assumes one of the two configurations from the diagram and never treats the other one.
- (NEC2) A solution that fails to exclude $n = 3$ (as this will throw off the indexing for the remainder of the proof and lead to an incorrect result since for odd indexes, no such x and z exist).
- (NEC3) A solution that states the relation $\frac{CP}{CH} = \frac{AH}{AQ}$ without proof or derives it only from a diagram-dependent, non-directed argument that fails in one configuration².
- (NEC4) A solution that gives only necessary conditions for n to be amazing, without justifying the converse.
- (NEC5) A solution that reduces to a Pell equation but applies the wrong parity or congruence class, thereby including the wrong Pell subsequence or missing every other amazing integer.
- (NEC6) A solution that obtains the correct Pell equations but does not determine the asymptotic behaviour of n_{2i} and hence does not determine both x and z .
- (NEC7) A solution that gives only a numerical guess for x , z , or the final answer, even if the guess is correct.

²Usually, diagram-dependence would not be penalised at the IMO; however, here having two configurations is integral to the problem so both must be dealt with.

Partial progress for incomplete solutions (0/1 marks)

Award 1 mark to an incomplete solution if it contains one of the following pieces of significant, correct progress:

(PP1) A valid proof of the core geometric relation

$$\frac{CP}{CH} = \frac{AH}{AQ},$$

with directed lengths or with a separate argument that correctly handles both configurations.

(PP2) A correct reduction from the geometry and integrality conditions to the two equations

$$\begin{cases} u^2 = 3m^2 - 3m + 1, \\ v^2 = 3m^2 + 3m + 1, \end{cases}$$

including the fact that $n = 3m^2$ with m odd and the correct interpretation of the two signs.

Do not award the partial-progress mark for any of the following on its own:

- (NPP1) Setting up coordinates for the geometry, or computing some of the points B, C, H, X, H_C , without reaching a substantive relation among CP, CH, AH , and AQ .
- (NPP2) Stating, guessing, or reading from the diagram that $\frac{CP}{CH} = \frac{AH}{AQ}$, without a valid proof.
- (NPP3) Deriving only that $n = 3m^2$, without obtaining the two square conditions or an equivalent Pell reduction.
- (NPP4) Solving the generic Pell equation $N^2 - 3M^2 = 1$ without connecting the solutions to the correct parity and congruence classes required by the problem.
- (NPP5) Producing numerical examples of amazing integers, or correctly guessing the final answer, without a proof of the relevant structure.
- (NPP6) Calculating x or z from an asserted sequence of amazing integers, unless the asserted sequence has been justified by one of the substantial reductions above.